A Comparative Study of Transcript-Only and Structured Caption Generation

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Abstract

Generating social media captions from conversational audio is challenging because transcripts often capture spoken words but miss important conversational signals such as emotion, intent and subtext. This project investigates whether incorporating structured conversational signals can improve caption generation compared with transcript-only methods. A modular system was developed in which short audio clips were first transcribed using OpenAI Whisper. Two caption-generation approaches were then compared: (1) a baseline method that generated captions directly from transcripts and (2) a structured method that first extracted conversational signals such as emotion, intent, themes and intensity, which were then used to guide caption generation. The two approaches were evaluated through a blinded human study using 50 conversational audio clips and five evaluators. Captions were assessed using relevance, emotion alignment, subtext and authenticity, along with overall preference selection, following the use of human-centered evaluation for open-ended text generation tasks. Results show that the structured approach achieved higher scores across all metrics, with the largest improvement in emotion alignment (5.740 vs. 5.028). Evaluators also preferred the structured approach in 64.8% of comparisons, compared with 35.2% for the baseline method. These findings suggest that incorporating structured conversational signals can improve caption quality, particularly in representing emotional tone, while transcript-only methods remain competitive for general relevance.

1 Introduction

Social media captions are a common way people share thoughts, emotions and experiences in a short and engaging format. However, writing meaningful captions is not always easy, especially when the ideas come from informal spoken conversations. In everyday conversations, people often express reflections, opinions and emotional moments that could be turned into captions. But converting these spoken ideas into written captions requires more than just transcribing the words; it also requires capturing the emotional tone, intent and underlying meaning of the conversation. Speech-to-text systems can convert spoken audio into text [4], but transcripts alone may miss important conversational signals such as emotion, intent and subtext. Because of this, captions generated from transcripts alone may feel generic or may not fully reflect the original conversation. Prior research has also shown that emotional meaning can be inferred from text, which suggests that transcripts may contain useful signals beyond literal words [3, 6]. To address this limitation, this project explores whether incorporating structured conversational signals can improve caption generation from conversational audio.

Research Question: How can an audio transcript generator using captured emotions, intent and themes improve AI-generated captions?

This question focuses on whether incorporating structured conversational signals, specifically emotion, intent and themes, provides additional context that improves caption generation compared to relying on transcript-only input. The study evaluates whether these signals lead to captions that better reflect the tone and meaning of the original conversation. This is assessed through human evaluation across relevance, emotion alignment, subtext and authenticity, as well as overall preference [1].

To answer this question, the project compares a baseline transcript-only approach with a structured tone-guided approach using human evaluation based on relevance, emotion alignment, subtext and authenticity.

The objectives of this project are to (1) implement a system that converts conversational audio into captions, (2) incorporate structured conversational signals into caption generation and (3) evaluate the impact of these signals through a controlled human study.

2 Background

Recent advances in machine learning have made it possible to process conversational audio and generate useful text from it. This project builds on three main areas: speech-to-text transcription, language model-based text generation and emotion detection from text. Speech-to-text systems, such as OpenAI's Whisper [4], are widely used to convert audio into text. These systems perform well on conversational speech and can handle informal language and different speaking styles. However, they focus on capturing what is said and do not explicitly represent emotional tone or intent clearly.

Large language models (LLMs) are commonly used for generating text, including captions. In many cases, captions are generated directly from transcripts. While this approach can capture the main idea of a conversation, it relies on the model to infer tone and meaning on its own. As a result, the generated captions may miss emotional nuance or deeper meaning. Prior work in automated audio captioning has also shown growing interest in generating natural language descriptions from audio inputs [2]. There has also been research on detecting emotion and sentiment from text [3, 6]. These methods can identify emotional signals, but they are usually treated as separate tasks and are not directly used to guide text generation. More recent work in prompt design suggests that giving structured information to language models can improve output quality [5]. Based on this, there are two main limitations. First, transcript-based caption generation does not explicitly include emotional or conversational signals. Second, emotion detection is often not integrated into the generation process in a structured way. In addition, evaluating generated captions often requires human judgment because qualities such as tone, authenticity and relevance can be difficult to measure automatically [1].

This project addresses these limitations by incorporating structured conversational signals into caption generation and evaluating their impact through human evaluation.

3 Approach

The project is developed using a modular pipeline that transforms conversational audio into captions. A Streamlit interface allows users to upload short audio clips. These clips are transcribed using OpenAI Whisper to produce a text transcript [4]. Two caption-generation pipelines are used. The baseline pipeline generates captions directly from the transcript. The structured pipeline first performs text-based conversational signal inference that produces a constrained representation containing signals such as primary emotion, conversational intent and themes. This structured representation is then used to guide caption generation.

The system is implemented using Python due to its strong support for machine learning and natural language processing libraries. Streamlit is used to provide a simple interface for uploading and testing audio inputs. OpenAI Whisper is selected for transcription due to its robustness to conversational speech and the GPT-4o-mini model is used for caption generation to provide consistent and controlled outputs. The evaluation uses a dataset of 50 short conversational audio clips collected from YouTube. The clips are approximately one minute in length and include conversational and reflective speech. Each audio clip is transcribed once and reused across both baseline and structured caption generation approaches to ensure consistency during evaluation. This design allows a direct comparison between transcript-only and structured approaches, enabling the evaluation of whether incorporating conversational signals improves caption quality. Figure 1 illustrates the overall system architecture and flow of the caption generation pipeline.

3.1 Speech-To-Text Transcription

As shown in Figure 1, the first stage of the system converts conversational audio into text using the Whisper model [4]. In this implementation, the "base" variant of the model is used to provide a balance between transcription accuracy and computational efficiency. Given an input audio file, the system applies Whisper's transcription function to generate a textual representation of the spoken content. The output of this stage is a plain text transcript, which is then used as the input for caption generation. A key design decision in this stage is that the same transcript is reused for both the baseline and structured caption generation approaches. This ensures that any differences in generated captions are due to the caption generation method rather than variations in transcription, thereby maintaining consistency in the evaluation. The use of Whisper is particularly suitable for this project because conversational audio often includes informal speech patterns, pauses and variations in tone. Whisper is robust to such variations, making it effective for capturing real-world conversational content.

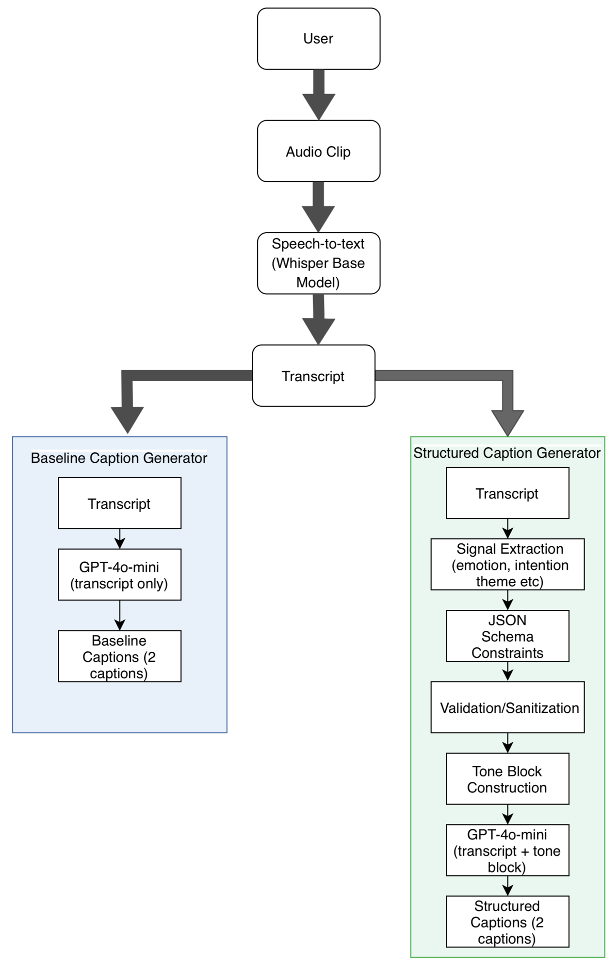

Figure 1: System architecture for transcript-only and structured signal-based caption generation.

3.2 Baseline Caption Generation

In the baseline approach, captions are generated directly from the transcript without incorporating any additional conversational or emotional signals. The transcript produced by the speech-to-text component is provided as the sole input to a language model (GPT-based), which generates caption suggestions based only on the textual content.

The system is designed to return exactly two captions for each transcript:

- one reflective caption (1–3 sentences)
- one short, punchy caption (1 sentence)

To ensure consistency and improve output quality, the prompt includes specific constraints. The model is instructed to produce concise, natural captions that are directly grounded in the transcript. Generic motivational phrases and cliches are explicitly discouraged to maintain authenticity and relevance to the original conversation.

This baseline method serves as the control condition in the evaluation. Because it relies only on the transcript, it does not explicitly account for emotional tone, conversational intent, or underlying subtext. As a result, while it may capture the main idea of the conversation, it may fail to fully reflect nuanced emotional or contextual aspects of the spoken content.

The following example illustrates the baseline caption generation approach using one transcript from the evaluation dataset. **Transcript**: *Some of the most successful people in the world are the ones who've had the most failures. J.K. Rowling, who wrote Harry Potter. Her first Harry Potter book was rejected 12 times before it was finally published. Michael Jordan was cut from his high school basketball team. He lost hundreds of games and missed thousands of shots during his career. But he once said, I have failed over and over and over again in my life. And that's why I succeed. These people succeeded because they understood that you can't let your failures define you. You have to let your failures teach you. You have to let them show you what to do differently the next time.*

Using only the transcript as input, the baseline method generated the following captions:

Caption 1: *Success isn't about never failing; it's about learning from those failures. From J.K. Rowling's countless rejections to Michael Jordan's missed shots, it's clear that setbacks can shape a path to achievement.*

Caption 2: *Failures are just stepping stones to what's next.* This example shows that the baseline approach can capture the central message of the transcript using only textual content, but it relies entirely on implicit cues present in the transcript.

3.3 Structured Caption Generation

The structured approach extends the baseline method by introducing an intermediate conversational signal extraction stage before caption generation. Instead of generating captions directly from the transcript, the system first analyzes the transcript to produce a structured representation of tone and meaning. This representation includes multiple conversational signals such as primary emotion, secondary emotions, intensity, sarcasm likelihood, confidence, intent, themes and supporting evidence. These signals are generated using a language model in a text-only setting, meaning the analysis relies entirely on the transcript rather than acoustic features, consistent with prior work on emotion detection from text [3, 6].

A central design feature of this approach is the use of a schema-constrained JSON representation. Rather than allowing the model to produce free-form output, the system enforces a predefined schema that specifies required fields, valid data types and structural constraints. This ensures that every output follows a consistent format across all transcripts.

The use of a schema provides several important benefits. First, it standardizes the structure of the intermediate representation, allowing the system to process outputs reliably in subsequent stages. Second, it reduces variability in model responses by preventing the use of inconsistent or invented labels. For example, emotions are restricted to a predefined set of allowed categories and intent labels must belong to a fixed list of conversational intents. Third, it improves comparability across samples, which is essential for evaluation, as all structured outputs follow the same format. Additional constraints are applied within the schema to further improve consistency. For example, the number of secondary emotions and themes is limited, intensity values are restricted to a bounded range and sarcasm likelihood is selected from a fixed set of options. These constraints reduce ambiguity and ensure that the structured representation remains compact and interpretable.

Overall, the schema transforms the conversational signal extraction stage into a controlled intermediate representation, making the structured pipeline more reliable and consistent compared to unconstrained text-based analysis. The structured output generated by the schema-constrained model is not used directly without verification. A validation and sanitization step is applied to ensure that all fields conform to the expected format and constraints. This step corrects invalid or inconsistent values, such as replacing unsupported emotion labels with default values, removing duplicate entries, clamping numerical ranges and enforcing limits on list sizes. In cases where the model output cannot be parsed correctly, a safe fallback representation is used. This ensures that the system remains stable and continues to function even when the model produces imperfect outputs. The sanitization process plays a critical role in maintaining consistency across the dataset and preventing errors in downstream processing.

Once validated, the structured representation is converted into a formatted intermediate representation, referred to as a tone block. This tone block summarizes the extracted conversational signals in a structured textual format and is provided as additional context to the caption generation model. The caption generation stage then uses both the original transcript and the tone block as input. This allows the model to incorporate explicit information about emotion, intent and thematic content when generating captions. As in the baseline approach, the system generates exactly two captions: one reflective caption consisting of 1–3 sentences and one short, punchy caption consisting of a single sentence. A fixed and low temperature setting is used during generation to ensure stable and consistent outputs across different samples. This is important for evaluation, as it minimizes variability that is unrelated to the structured representation itself.

By combining schema-constrained extraction, validation and tone-guided generation, the structured pipeline introduces a controlled mechanism for incorporating conversational signals into caption generation. This design aims to produce captions that better reflect emotional tone, contextual meaning and underlying subtext compared to transcript-only methods.

The same transcript from Section 3.2 is used in the structured pipeline, but the system first extracts conversational signals before generating captions.

The structured analysis identified the following signals:

Primary Emotion: hope and optimism

Secondary Emotions: joy, frustration

Intensity: moderate

Intent: motivational statement

Themes: success, failure, learning, resilience

Using these extracted signals together with the transcript, the structured method generated the following captions:

Caption 1: *Every setback is a lesson in disguise. Just look at J.K. Rowling and Michael Jordan—failure was just a stepping stone on their paths to greatness. Embrace the lessons and let them guide you forward.*

Caption 2*: Failures are just the plot twists in your success story!*

This example demonstrates how the structured method uses extracted conversational signals to guide caption generation toward stronger emotional framing and thematic consistency.

4 Evaluation

To compare the baseline and structured caption generation approaches, a controlled human evaluation is conducted using a dataset of short conversational audio clips. The overall evaluation workflow is shown in Figure 2.

The dataset consists of 50 conversational audio clips. Each clip is transcribed using the speech-to-text component of the system. For each transcript, two sets of captions are generated: one using the baseline transcript-only approach and one using the structured tone-guided approach.

During evaluation, a total of five evaluators participated in the evaluation process. Each evaluator first listens to the original audio clip to understand the context and tone of the conversation. The evaluator then reads the transcript, followed by both caption sets.To ensure a fair comparison, the captions are presented in a blinded format. For each clip, the two caption sets are labeled as "Set X" and "Set Y," and their order is randomized. This prevents evaluators from knowing which method produces each caption and reduces potential bias. Evaluators assess each caption set independently based on four criteria: relevance, emotion alignment, subtext and authenticity. Each criterion is scored on a scale from 1 to 7 using a defined rubric. In addition to scoring, evaluators select which caption set (X or Y) they prefer overall. This provides a direct comparison between the two approaches and helps identify which method produces more effective captions in practice. Human evaluation is particularly useful for language generation tasks where qualities such as tone, relevance and authenticity can be difficult to measure automatically [1].

After evaluation, the collected scores are aggregated to compare the baseline and structured approaches. Since evaluators assess caption sets labeled as X and Y, a mapping is used to convert these labels back to their corresponding methods (baseline or structured). For each approach, the scores across all clips and evaluators are averaged for each metric (relevance, emotion alignment, subtext and authenticity). These average scores provide a quantitative comparison of how each method performs across different aspects of caption quality.

In addition to metric averages, preference selections are also aggregated by counting how many times each approach is chosen by evaluators. This provides a direct measure of which method is more often preferred in practice. The structured approach is considered to perform better if it achieves higher average scores across the evaluation metrics and receives a higher number of preference selections compared to the baseline approach.

To ensure consistency in scoring, a rubric is defined for each evaluation metric. Scores are interpreted as follows: 1–2 indicates poor performance, 3–4 indicates moderate performance, 5–6 indicates good performance and 7 indicates excellent performance. Each metric (relevance, emotion alignment, subtext and authenticity) is further supported by example-based descriptions that illustrate how different score levels correspond to caption quality. These examples help evaluators apply the scoring criteria consistently across different audio clips.

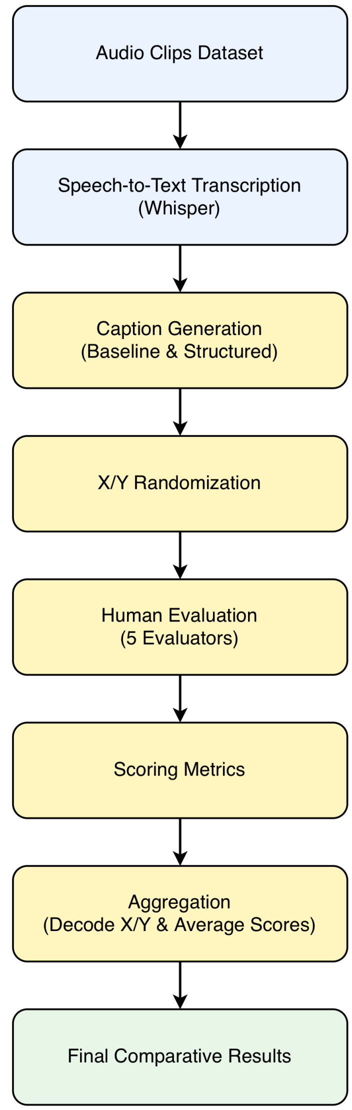

Figure 2: Evaluation workflow for comparing baseline and structured caption generation outputs.

4.1 Evaluation Rubric

To guide evaluators and ensure consistent scoring, an example-based rubric is used for each evaluation metric. Each metric is scored on a scale from 1 to 7, where 1–2 indicates poor performance, 3–4 indicates moderate performance, 5–6 indicates good performance and 7 indicates excellent performance.

Sample Transcript: "*I thought I'd be fine with it, but honestly it still stings a little. Maybe I just need more time.*" Relevance measures how well the caption reflects the main idea of the conversation. Scores of 1–2 correspond to captions that are unrelated to the conversation, such as "*Living my best life every day.*" Scores of 3–4 indicate partial alignment with the topic but lack specificity, for example, "*Sometimes things don't go as planned.*" Scores of 5–6 represent captions that are mostly relevant but may miss some nuance. A score of 7 indicates that the caption clearly captures the main idea of the conversation, such as "*I thought I'd be okay with it, but it still hurts more than I expected.*"

Emotional Alignment measures how well the caption matches the emotional tone of the conversation. Scores of 1–2 represent captions with completely mismatched emotions, such as "*Feeling amazing and grateful today!*" Scores of 3–4 reflect partial emotional alignment, for example, "*Things have been a bit tough lately.*" Scores of 5–6 indicate generally correct emotional tone with minor inconsistencies. A score of 7 reflects strong emotional consistency with the original speech, such as "*It still stings a little, even though I try to move on.*"

Subtext measures how well the caption captures implied meaning beyond literal words. Scores of 1–2 correspond to captions that simply restate content without interpretation, such as "*I talked about something that happened.*" Scores of 3–4 reflect some level of implied meaning, for example, "*Maybe I just need more time.*" Scores of 5–6 indicate moderate interpretation of underlying meaning. A score of 7 reflects deeper emotional nuance and implicit meaning, such as "*I keep telling myself I'm fine, but I'm not fully over it.*"

Authenticity measures how natural and suitable the caption is for social media. Scores of 1–2 correspond to captions that sound unnatural or overly formal, such as "*This situation has contributed to my emotional processing.*" Scores of 3–4 represent captions that are acceptable but not particularly engaging, for example, "*Today was a bit difficult.*" Scores of 5–6 indicate generally natural captions with minor issues. A score of 7 reflects captions that feel natural, relatable and appropriate for social media, such as "*Some things just take longer to heal than you expect.*" By grounding each metric in explicit score ranges and concrete examples, this rubric provides evaluators with clear reference points for interpreting scores. This reduces subjectivity and helps ensure consistency across different evaluators and audio clips.

5 Results

This section presents the results of the human evaluation comparing the baseline transcript-only approach and the structured tone-guided approach.

Table 1: Average evaluation scores across metrics.

Metric	Baseline	Structured
Relevance	5.124	5.160
Emotion Alignment	5.028	5.740
Subtext	4.632	4.716
Authenticity	4.720	4.744

As shown in Table 1, the structured approach achieves higher scores across all four-evaluation metrics. The largest improvement is observed in emotion alignment, where the structured method scores 5.740 compared to 5.028 for the baseline approach, indicating a clear advantage in capturing the emotional tone of the conversation. Smaller improvements are observed in relevance and subtext. Relevance scores are similar between the two approaches (5.160 vs 5.124), suggesting that transcript-only input is generally sufficient for capturing the main idea. Subtext shows a modest increase (4.716 vs 4.632), indicating some improvement in capturing implied meaning. Authenticity scores remain nearly identical, suggesting that both approaches produce similarly natural captions. In addition to metric-based evaluation, preference selections were aggregated to directly compare the two approaches.

Table 2: Preference comparison between baseline and structured approaches.

Approach	Preference Count	Percentage
Baseline	88	35.2%
Structured	162	64.8%

As shown in Table 2, out of 250 evaluator–clip comparisons, the structured approach is preferred in 162 cases (64.8%), while the baseline approach is preferred in 88 cases (35.2%). This indicates that evaluators more frequently favored captions generated using structured conversational signals. This preference trend reinforces the metric-based results, particularly the improvement observed in emotion alignment. To provide transparency into the evaluation process, a representative sample of the raw evaluation scores is shown in Table 3. The metric abbreviations are as follows: Rel. = Relevance, Emo. = Emotion Alignment, Sub. = Subtext, Auth. = Authenticity, and Pref. = Preference.

Table 3: Sample raw evaluation scores for one audio clip.

Clip	Eval.	Set	Rel.	Emo.	Sub.	Auth.	Pref.
audio_1	E1	X	5	5	5	5	Y
audio_1	E1	Y	6	6	5	5	Y
audio_1	E2	X	5	5	4	5	X
audio_1	E2	Y	6	7	5	6	X
audio_1	E3	X	6	6	6	6	X
audio_1	E3	Y	5	6	5	5	X
audio_1	E4	X	4	4	3	4	X
audio_1	E4	Y	4	5	4	4	X
audio_1	E5	X	6	6	5	5	X
audio_1	E5	Y	7	7	7	6	X

Table 3 illustrates how individual evaluators scored both caption sets across multiple metrics, demonstrating the variability and consistency in scoring across different evaluators and clips. Overall, the results show that incorporating structured conversational signals leads to improvements in caption quality, particularly in capturing emotional tone and is more frequently preferred by human evaluators.

6 Discussion

The results suggest that incorporating structured conversational signals improves caption generation in specific areas, particularly in capturing emotional tone. The structured approach achieved a clear improvement in emotion alignment compared to the baseline method, indicating that explicitly providing emotion, intent and thematic information help guide the model toward outputs that better reflect the tone of the original conversation. In contrast, the baseline approach relies on implicit cues within the transcript, which may not always convey emotional context effectively. However, improvements were not consistent across all metrics. The structured approach showed only small differences in relevance, subtext and authenticity. This suggests that transcript-only inputs are already sufficient for capturing the general content of the conversation and that structured signals primarily enhance the model's ability to represent emotional context rather than overall content. Subtext and authenticity require deeper contextual understanding and nuance, which may not be fully captured through the current structured representation.

The preference results further support this observation, as evaluators selected the structured captions more frequently. This indicates that improvements in emotional alignment have a noticeable impact on how captions are perceived, even when other metrics show only minor differences. Based on these findings, the research question is addressed by demonstrating that incorporating structured conversational signals improves AI-generated captions, especially in terms of emotional representation. While the improvements are not uniform across all aspects of caption quality, the consistent preference for structured captions suggests that emotional alignment plays a key role in perceived effectiveness. These findings suggest that incorporating structured conversational signals can improve the effectiveness of caption generation systems, particularly in applications where emotional tone is important, such as social media content creation. The use of multiple evaluators and a blinded evaluation setup further support the reliability of these results, as the observed improvements are consistent across different evaluators and audio clips.

One limitation of this work is that the structured signals are derived only from text transcripts and do not include acoustic features such as tone, pitch, or pauses in speech. Additionally, the dataset size is limited and results may vary with a larger or more diverse set of audio inputs. Another limitation is that the audio clips used for evaluation are relatively short, with most clips being approximately one minute in duration. While this helps maintain consistency and simplifies evaluation, it may limit the ability to capture longer conversational context and evolving emotional patterns. Future work could explore incorporating acoustic features, expanding the dataset and using longer and more diverse audio inputs to further evaluate the impact of structured conversational signals.

7 Conclusion

This project studies how adding structured signals such as emotion, intent and themes can improve AI-generated captions from conversational audio. A system was built to generate captions using two approaches: a transcript-only baseline and a structured tone-guided method. The results show that the structured approach improves captions mainly in terms of emotional alignment, while changes in relevance, subtext and authenticity are small. Evaluators also preferred structured captions more often. This suggests that structured signals help capture the tone of a conversation, but their overall impact is limited to certain aspects. Future work can explore using audio features such as tone and pitch and testing the approach on larger and more diverse datasets.

AI Use Disclosure

AI was used to assist with refining the structure of the paper and improving clarity of wording. AI suggestions were limited to rephrasing, organization, and formatting support. AI was not used to generate the research question, evaluation results, or analysis. All content was reviewed and revised to ensure that it accurately reflects the implemented system, evaluation process, and findings of the project.

References

[1] Chia-Wei Liu, Ryan Lowe, Iulian V. Serban, Michael Noseworthy, Laurent Charlin, and Joelle Pineau. 2016. How NOT To Evaluate Your Dialogue System: An Empirical Study of Unsupervised Evaluation Metrics for Dialogue Response Generation. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*. doi:10.18653/v1/D16-1230

[2] Xinhao Mei, Xubo Liu, Mark D. Plumbley, and Wenwu Wang. 2022. Automated Audio Captioning: An Overview of Recent Progress and New Challenges. *EURASIP Journal on Audio, Speech, and Music Processing* (2022). doi:10.1186/s13636-022-00259-2

[3] Pansy Nandwani and Rupali Verma. 2021. A Review on Sentiment Analysis and Emotion Detection from Text. *Social Network Analysis and Mining* (2021). doi:10.1007/s13278-021-00776-6

[4] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2022. Robust Speech Recognition via Large-Scale Weak Supervision. *arXiv preprint arXiv:2212.04356* (2022). doi:10.48550/arXiv.2212.04356

[5] Pranab Sahoo, Ayush Kumar Singh, Sriparna Saha, Vinija Jain, Samrat Mondal, and Aman Chadha. 2024. A Systematic Survey of Prompt Engineering in Large Language Models: Techniques and Applications. *arXiv preprint arXiv:2402.07927* (2024). doi:10.48550/arXiv.2402.07927

[6] Samira Zad, Maryam Heidari, James H. Jr. Jones, and Ozlem Uzuner. 2021. Emotion Detection of Textual Data: An Interdisciplinary Survey. In *2021 IEEE International Conference on Artificial Intelligence and Internet of Things (AIIoT)*. doi:10.1109/AIIoT52608.2021.9454192